Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CT5DE12/18MA5DE12]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

SYSTEM SOFTWARE

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Develop the SIC program to implement the following arithmetic expression BETA = ALPHA +1. **06**

b. Compare SIC and SIC/XE machine in the context of Instruction set, Addressing modes, memory, Instruction format and data format. **08**

2. a. Consider the memory locations and its contents shown in Table-1 and the present relative value of Program Counter (PC), Base (B) and index register are 003000, 006000 and 000090 respectively. **08**

Table-1

MEMORY LOCATION	CONTENT
3030	003600
3600	103000
6390	00C303
C303	003030

If the bit positions of n,i,x,b,p,e and disp/address are

n	i	x	b	p	e	disp/address
1	1	0	0	1	0	0110 0000 0000
1	1	1	1	0	0	0011 0000 0000
0	1	0	0	0	0	0000 0011 0000
1	0	0	0	1	0	0000 0011 0000

Compute the Target address and the value to be loaded in register A.

b. **Use of all machine dependent features (SIC/XE features), convert the the following program to SIC/XE Program.** **06**

LINE.NO	LABEL	MNEMONIC	OPERAND
1	SUM	START	1000
2	FIRST	LDA	ZERO
3		LDX	ONE
4	LOOP	STX	I
5		ADD	I
6		TIX	ELEVEN
7		JLT	LOOP
8	ONE	WORD	1
9	I	RESW	1
10	ZERO	WORD	0
11	ELEVEN	WORD	11
12		END	FIRST

3.	**a.**	Describe about the machine dependent features of loader in detail.	**14**
		OR	
4.	**a.**	Explain the Absolute Loader and write its algorithm.	**14**
5.	**a.**	Express the basics of macro definition and macro expansion with an example.	**10**
	b.	Analyze the contents of ARGTAB for the following macro invocation. **Copy F1,F2,S1,S2**	**04**
		OR	
6.	**a.**	Illustrate the implementation of conditional macro expansion statements depending on the arguments supplied in macro invocation.	**14**
7.	**a.**	Describe the structure of LEX Program and Explain the work flow of the LEX complier	**07**
	b.	Write a Lex program that finds the longest word (defined as a contiguous string of upper and lower case letters) in the input.	**07**
		OR	
8.	**a.**	Write the YACC Program to recognize the string „abbb", „ab" „a" of the language (a^nb^n , n>=1).	**07**
	b.	Explain the structure of YACC Program and the work flow of the YACC complier	**07**